



JAIDEV EDUCATION SOCIETY'S  
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(An Autonomous Institute, with NAAC "A" Grade)  
Affiliated to DBATU & RTMNU



**B. Tech Summer / Winter - 2025**  
**(ESE / Make-up / Supplementary) Examination**

**Branch: Computer Science & Engineering**

**Sem:- V**

**Subject: Database Management Systems**

**Subject Code: CSST001**

**Marks: 100**

**Date of Exam: 27/10/2025**

**Time: 3 Hrs.**

**Instructions to the students:**

1. Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.
2. All the questions carry marks as indicated.
3. Assume suitable data whenever necessary.
4. Illustrate your answer whenever necessary with the help of neat sketches.
5. Use of Programmable Calculator is prohibited.
6. Don't use red pen for writing the answers.

**COs: Students should be able to:**

1. Understand fundamentals of database management system.
2. Classify the different query development knowledge.
3. Apply modelling and normalization to databases.
4. Analyse query processing and optimization techniques.
5. Evaluate the File Organization, Indexing and exhibit the knowledge of transaction and concurrency control.

**SECTION A**

1. All Questions are compulsory.
2. Answer in one/two word/sentence.

| Q. No. | Questions                                                                      | Level/CO<br>Mapped | Marks |
|--------|--------------------------------------------------------------------------------|--------------------|-------|
| Q.1    | Define the term 'schema'.                                                      | L1/ CO1            | 2     |
| Q.2    | State the primary goal of the Normalization process.                           | L1/CO3             | 2     |
| Q.3    | Identify the normal form that addresses transitive dependency.                 | L2/CO3             | 2     |
| Q.4    | List out the three main levels of Data Abstraction.                            | L1/CO1             | 2     |
| Q.5    | Rewrite the relational algebra expression for the Rename operation ( $\rho$ ). | L2/CO2             | 2     |
| Q.6    | Classify the different types of Join operations in SQL.                        | L2/CO2             | 2     |
| Q.7    | Restate the purpose of the Rollback command.                                   | L2/CO5             | 2     |
| Q.8    | Write the DML command used to change data in a table.                          | L3/CO2             | 2     |
| Q.9    | Give an example of a functional dependency.                                    | L2/CO3             | 2     |
| Q.10   | Define the term 'transaction'.                                                 | L1/CO5             | 2     |
| Q.11   | State the purpose of Query Processing.                                         | L1/CO4             | 2     |
| Q.12   | Enlist two disadvantages of the File Processing System.                        | L1/CO1             | 2     |



## SECTION B

1. All Questions are compulsory.

| Q. No. | Questions                                                                            | Level/CO Mapped | Marks |
|--------|--------------------------------------------------------------------------------------|-----------------|-------|
| Q.13   | Distinguish between Tuple Relational Calculus and Relational Algebra.                | L4/CO2          | 4     |
| Q.14   | Illustrate the concept of Transitive Dependency with a non-normalized table example. | L3/CO3          | 4     |
| Q.15   | Discuss the concept of Data Abstraction in a DBMS.                                   | L2/CO1          | 4     |
| Q.16   | Examine the process of using B-trees for efficient data retrieval.                   | L4/CO5          | 4     |
| Q.17   | Write short note on the ACID Properties of a transaction.                            | L3/CO5          | 4     |
| Q.18   | Sketch the E-R Diagram for a simplified scenario (e.g., Department and Employee).    | L3/CO3          | 4     |

## SECTION C

1. Solve any FOUR Questions.

| Q. No. | Questions                                                                                                  | Level/CO Mapped | Marks |
|--------|------------------------------------------------------------------------------------------------------------|-----------------|-------|
| Q.19   | Illustrate the working of the Two-Phase Locking (2PL) protocol with a sample transaction schedule.         | L4/CO5          | 7     |
| Q.20   | Explain and justify the necessity of the Lossless Design property for database decomposition.              | L5/CO3          | 7     |
| Q.21   | Analyse the process of Query Optimization by decomposing a query plan into relational algebra expressions. | L4/CO4          | 7     |
| Q.22   | Examine the various File Organization methods.                                                             | L4/CO5          | 7     |
| Q.23   | Describe and critique the Timestamp-based Concurrency Control scheme.                                      | L5/CO5          | 7     |
| Q.24   | Discuss the differences between Relational Algebra and Relational Calculus.                                | L5/CO2          | 7     |

## SECTION D

1. Solve any TWO Questions.

| Q. No. | Questions                                                                                                                                  | Level/CO Mapped | Marks |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------|
| Q.25   | Rewrite about the following:<br>a) The concept of Serializability (Conflict and View)<br>b) The purpose of Database Recovery.              | L5/CO5          | 12    |
| Q.26   | Explain the concepts of Functional Dependencies and the different Normal Forms (2NF, 3NF, BCNF) with a detailed example of normalization.  | L6/CO3          | 12    |
| Q.27   | Describe the full process of Query Processing from parsing to execution, and construct a high-level overview of different Join Strategies. | L6/CO4          | 12    |